N83 E-1 full scan against zero gap: Tier 1 replay protocol

Status: run; verdict `promoted` (Tier 1, magnitude `refuted_overstated`). See Result. This directory froze a Tier 1 (cloud instance) paired replay of the n=83 Frobenius canonicalization stage, which settles the paper's exploratory canonicalization-phase ratio of **1.42**×. That figure (papers/frobenius-orbits/sec-benchmarks.tex, Table "survives") comes from three unisolated macOS runs per arm. This replay either promotes it to an instance-scoped Tier 1 result or refutes it.

Item	Value
Run	GitHub Actions run `38502816475`, `ubuntu-24.04`, instance `gh-38502816475-1-paired-stage`, receipt `tier1_github_38502816475/`
Proposal	`Q2026100902` (candidate ID `null`: only the PDP stage is specified)
Curve	`EC1N83Ckb1h876c2921cb64`
Workload	`981ac530e67e` (one frozen public target, 2,000,000 pair states, 1,024 MiB RSS cap)
Frozen experiment	../packed_index_e1_canonical_20261009/
Stage wrapper (reused unchanged)	../packed_index_e1_isolated_replay_20261010/benchmark_stage.py
Reference arm R	`packed_bucket22`: full 83-rotation scan
Candidate arm C	`packed_bucket22_fastcanon`: exact longest-zero-gap scan
Native source	`390f16b16800d0542bd0769e1d3f2ac80fdaa647e67184e23e8b586db71691cf`
Pair certificate	`7cfba6355dc0a0dd4c67bfcf585377b57caa50c4391554ac5ab5ec86d7acaff8`
Tier 1 protocol	tier1_protocol.json, SHA-256 `6ce0131883afb695f34efe9df90c6e38eb488c79675e5af4dabde6d5d0111b90`
Workflow	.github/workflows/n83-e1-canonical-tier1.yml

What is frozen

The Tier 1 protocol was written before any timing in this directory. It does the following:

- **Bindings.** It binds by SHA-256 the frozen canonical protocol, stage manifest, workload, summary and native source, plus the Cargo files, bridge, representatives, both expected raw rows, the stage wrapper and `scripts/isolated_bench.py`.
- **Pairs and order.** There are 16 same-target pairs in the order R C, C R, repeated eight times. That is the order `isolated_bench.py` produces for a single case. All arms use one release binary on one pinned CPU, with a 180 s cap per arm and a 5,760 s cap for the panel.
- **Correctness.** Each arm must reproduce every frozen arm field of its mode's published row. These fields include:
  - the operation counts: 328,000,000 / 656,000,000 rotations for the full scan and 19,127,078 / 38,261,354 for the zero-gap scan;
  - `lookup_sample`, which holds the canonical key and the stored Frobenius shift of the 64 roots of the first 32 indexed states.

  Both arms must also share the 20-field pair certificate.
- **Noise gates.** These are the Tier 1 preflight before the panel and before and after each arm, plus per-arm checks:
  - exit 0, no timeout, `verified=1`, and an unchanged executable;
  - zero `nr_throttled`, `oom` and `oom_kill` deltas;
  - native peak RSS of at most 1,024 MiB;
  - phase accounting: query prep + S3 batch + partner check ≤ target stage, and canonicalization + exact lookup ≤ partner check.

  Steal, IRQ, involuntary switches and child CPU time are recorded but not gated.

- **Statistics.** All three are paired ratios R/C, taken as the median of the 16 per-pair ratios. The 95% interval is a paired bootstrap: `random.Random(20261011)`, 10,000 resamples, sorted, indices 250 and 9749. This is the runner's own Tier 1 rule, applied with a fresh generator for each statistic.
  - *Primary*: canonicalization phase, `timing_ns.target_cycle_canonical`, decoded from each arm's embedded raw result.
  - *Secondary*: the target stage (`target_pdp_and_native_check`, the runner's `online_ms`).
  - *Negative control*: the S3 batch phase, which does identical work in both arms. It is reported to show instrument noise and is never gated.
- **Decision rule** (on the primary interval):
  - `invalid`: any gate fails. There is no verdict, and 1.42× stays exploratory.
  - `promoted`: the lower bound is above 1. The claim reports the measured median and interval, the provider, the instance and the tier. It does not report 1.42×.
  - `refuted_regression`: the upper bound is below 1.
  - `not_promoted_inconclusive`: the interval contains 1.
  - The magnitude check against 1.4232676 is separate: `consistent` / `refuted_overstated` / `refuted_understated`.
  - The secondary statistic is classified the same way but never changes the primary verdict.

  A promoted result is a stage diagnostic on one target and one instance. It is not a complete single-target IC speedup.

The exploratory 1.4232676 is a ratio of six-run medians (2,215,795,406 / 1,556,836,845 ns). The Tier 1 statistic is a median of paired ratios, so the two estimators differ, and the protocol records that.

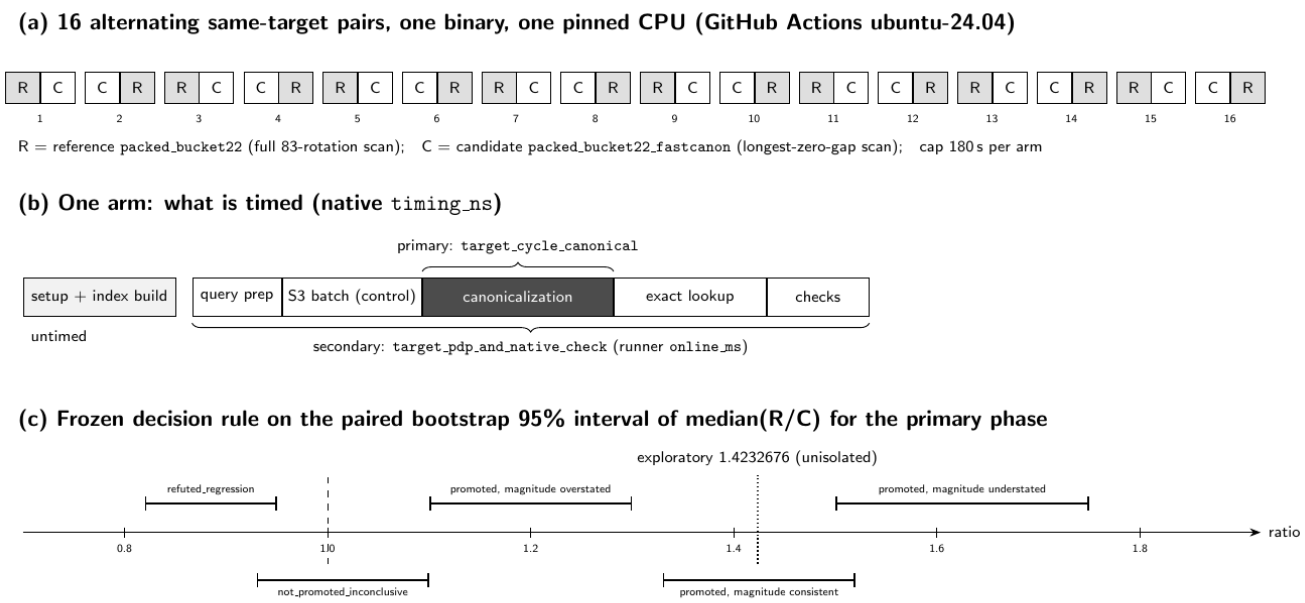


Figure 1: Frozen protocol: pair schedule, timed interval, and decision rule. Panel (b) is schematic: canonicalization and lookup are interleaved sub-intervals of the partner check, not consecutive blocks.

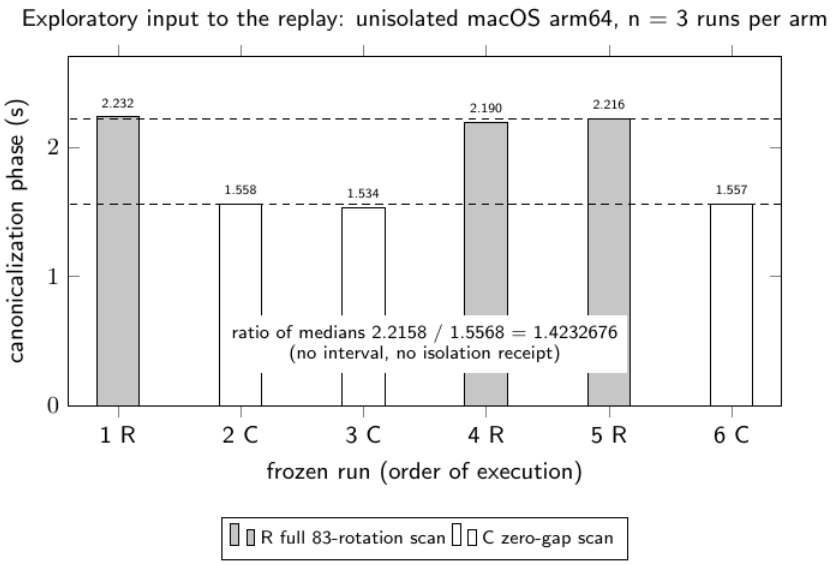


Figure 2: Exploratory input: the six frozen unisolated canonicalization-phase times (seconds) from the six raw rows of the frozen canonical experiment. No interval and no isolation receipt.

Files

File	Role
`tier1_protocol.json`	frozen pairs, order, gates, statistics, and decision rule
`verify_inputs.py`	checks every binding, the six frozen rows, the rotation counts, the pair certificate and the exploratory ratio
`restore_crypto_source.py`	restores crypto `904f0f84` into the frozen experiment's gitignored `crypto_source_snapshot/` and checks its tree digests
`make_tier1_manifest.py`	writes the `isolated_bench.py` Tier 1 manifest (sixteen pairs, one case)
`verify_tier1_receipt.py`	audits an archived receipt, replays the raw results, recomputes all ratios and intervals, and prints the verdict
`tier1_github_38502816475/`	archived receipt of the run: allocation, host probe, preflight, manifest, raw pairs, build log, `verdict.json`
`figures/*.tex`, `figures/*.pdf`, `figures/*.png`	diagram and graph TikZ sources, vector and PNG renderings
`README.pdf`	this report with its figures
`report_header.tex`, `report_filter.lua`	PDF build: `pandocREADME.md-oREADME.pdf--pdf-engine=pdflatex-Hreport_header.tex--lua-filterreport_filter.lua-Vgeometry:margin=2cm-Vfontsize=10pt-Vcolorlinks=true`

How it runs

A push to `claude/frobenius-orbit-canonicalization-gwmz8z` runs the workflow when the push touches the workflow or this directory. It can also be started with `workflow_dispatch`. The workflow does the following:

1. Records an allocation receipt and host probe for the fresh `ubuntu-24.04` runner.
2. Restores the pinned crypto source and runs `cargotest--release--locked`.
3. Builds the frozen binary.
4. Runs `verify_inputs.py`.

5. Writes the manifest and the Tier 1 preflight.
6. Runs the sixteen pairs.
7. Copies the runner output to `results/` and runs `verify_tier1_receipt.py` into `verdict.json`.
8. Uploads the whole receipt directory as artifact `n83-e1-canonical-tier1-<run_id>-<attempt>`, including failures.

To archive a receipt, download that artifact into `tier1_github_<run_id>/` here and run `python3verify_tier1_receipt.pytier1_github_<run_id>`.

Local preparation checks (2026-10-14, cloud agent VM; no timing claims)

These checks ran in a Firecracker VM inside Docker: 4 vCPU, Intel Xeon @ 2.10 GHz, rustc 1.97.0, Python 3.13.16. Other agents were running at the same time.

- **Build and tests.** `cargotest--release--locked` passed all 4 tests, including `zero_gap_canonical_matches_full_rotation_scan`, which is exhaustive at n = 3, 5, 7, 11 plus 100,000 fixed-seed 83-bit words. The release build produced binary SHA-256 `15690b8dc13ce8a9a18af4cc54ee1331b846b529038c17d3dd6070832e4f4a86` (Linux x86-64). Its digest is not the frozen macOS binary's and is not expected to be.
- **Input verification.** `verify_inputs.py` returned `PASS`.
- **Manifest.** `make_tier1_manifest.py` wrote a 16-repetition manifest with 446 hashed artifacts.
- **Probes.**
  - `isolated_bench.py probe` on the Tier 1 manifest returned `ok: true` with no problems. This host meets the Tier 1 preflight's mechanical checks (Linux, affinity, cgroup throttling and OOM counters). But it is a shared container, not the instance the protocol names, so the result is not used.
  - The Tier 0 manifest of the original adapter (`../packed_index_e1_isolated_replay_20261010/make_manifest.py`) was rejected with 16 problems, including Docker virtualisation, a missing isolated cgroup, `nohz_full`, frequency, `numactl`, and IRQ affinity.
- **Smoke run.** One unisolated arm of each mode ran through the frozen wrapper. Both reproduced every frozen arm field of the macOS rows on Linux x86-64.
  - Both returned `verified=1`.
  - The `lookup_sample` keys and shifts (64 roots) were identical, and so was the pair certificate.
  - Operation counts differ only in `canonical_rotation_steps`.

  Their timings are not reported. They come from an unqualified host.
- **Receipt verifier.** To exercise `verify_tier1_receipt.py`, the full 16-pair panel ran once through `isolated_bench.py run` on this shared VM, using a local allocation receipt (`local_agent_vm`). The panel completed: 16/16 valid pairs, artifacts stable. The verifier then replayed it without error and returned a verdict.
  - Setting one arm's `nr_throttled` delta to 1 gave verdict `invalid`.
  - Corrupting one arm's raw-result digest raised an error.

  The panel is a code-path test only. It is not committed, and its timings and verdict are not a measurement.

Result

Run `38502816475` (commit `66e16796`, public runner image `ubuntu24` 20261004.327.1, 4 exposed CPUs, one pinned) completed with conclusion `success`. The artifact was downloaded through the GitHub API and archived in `tier1_github_38502816475/` without the runner's queue lock file. `verify_tier1_receipt.py` replayed the archived raw results from the committed copy and reproduced the workflow's `verdict.json` byte for byte. All 16 pairs were valid and every gate passed (peak native RSS 562,561,024 bytes against the 1,024 MiB cap).

Statistic (paired R/C, median of 16)	Median	95% paired bootstrap	Rule outcome
Primary: canonicalization phase	1.0856	[1.0757, 1.0939]	`promoted` (lower bound above 1)
Secondary: target stage	1.0227	[1.0174, 1.0327]	`promoted`, never changes the primary
Negative control: S3 batch phase	1.0073	[0.9968, 1.0171]	not gated; interval contains 1

Median canonicalization phase: 1,172,834,420 ns (R) against 1,079,689,756 ns (C). Median target stage: 4,419.0 ms (R) against 4,272.3 ms (C).

Verdict by the frozen rule. `promoted`: the primary lower bound is above 1. The claim is a canonicalization-phase ratio of 1.086 [1.076, 1.094] on GitHub Actions, instance `gh-38502816475-1-paired-stage`, tier 1. The magnitude check is `exploratory_magnitude_refuted_overstated`: the exploratory 1.4233 lies far outside the interval, so 1.42x is not supported and is not reported as a result. The negative control stayed near 1 as intended.

Limits: one target, one instance, a stage diagnostic (canonicalization is about a quarter of the target stage; the target stage moves only 1.023x). A container affinity mask does not prove host-wide exclusivity, and steal and involuntary switches were recorded, not gated. This is not a complete single-target IC speedup and is not a Tier 0 result.

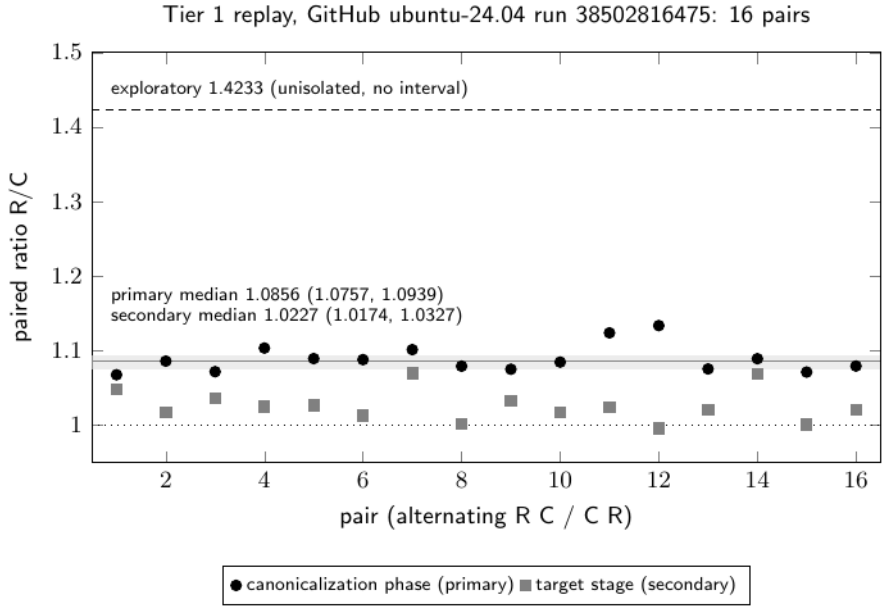


Figure 3: Paired R/C ratio of each of the 16 alternating pairs, with the median and 95% interval of the primary statistic and the exploratory figure for scale.

Graphs checked

`figures/tier1_result.{tex,pdf,png}` is new and plots the receipt. The paper's Table "survives" row and the `sec-benchmarks.tex` zero-gap paragraph now cite the Tier 1 figure and state that the unisolated 1.42x is refuted; the frozen-protocol and exploratory-input figures are unchanged and remain current.